

Vessel-Type Classification from NOAA AIS Using a Compact Feature Set and a Residual Feed-Forward Network

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Fundamentals of data science / Sapienza

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Abstract

Automatic Identification System (AIS) records are central to maritime monitoring, yet the ship-type field can be missing, noisy, or deliberately falsified, limiting downstream analytics and situational awareness. We build an end-to-end supervised pipeline to predict vessel macro-classes from a ship-level dataset derived from NOAA AIS data (via Kaggle), targeting reliable classification with minimal inputs. Each vessel (MMSI) is represented by **11 numerical features** capturing geometry and simple kinematics: length, width, draft, dimensionless ratios (e.g., L/W , L/D), footprint proxies (area and log-area), and speed statistics (mean SOG and log-mean SOG). To address severe class imbalance across **17 classes**, we adopt a hybrid strategy combining capped undersampling of majority classes, conservative oversampling of rare classes, and a class-weighted loss. A residual feed-forward neural network trained on standardized features achieves approximately **0.81 test accuracy**, with errors concentrated among operationally similar categories. The resulting baseline is fast, interpretable at the feature level, and suitable as a reliable building block for richer AIS behavior models.

1 Introduction

AIS streams provide global-scale vessel information (static metadata and dynamic navigation signals), enabling applications such as traffic monitoring, anomaly detection, and maritime safety. However, AIS self-reported attributes (especially vessel type) can suffer from errors and manipulation, motivating automated ship-type inference from robust descriptors. Prior work suggests that combining geometric characteristics with behavior-related features improves classification performance compared to using either alone. We implement a compact, reproducible baseline: (i) engineer a small ship-level feature vector, (ii) mitigate imbalance with hybrid sampling + class-weighted loss, and (iii) train a residual feed-forward neural network optimized for tabular data.

2 Data and Features

2.1 Source and unit of analysis

Starting from NOAA AIS data (distributed through Kaggle), we construct a **ship-level** dataset with one row per MMSI, aligning static ship metadata with aggregated dynamics (speed statistics). This yields **MMSI, label, and 11 features** (plus bookkeeping columns in earlier stages).

2.2 Feature set (11D)

Let L =length, W =width, D =draft. We use:

- **Geometry / shape:** $L, W, D, L/W, W/L, L/D, W/D$.
- **Scale proxy:** $area_proxy = L \cdot W$, and $\log(area_proxy)$ to reduce skew.
- **Dynamics:** sog_mean and $\log(sog_mean)$ (speed is often heavy-tailed).

The target label spans **17 macro-classes** (Cargo, Fishing, Passenger, Tanker, Tug, Dredging / Underwater ops, Diving ops, Military ops, Sailing and Pleasure, Pilot Vessel, Search and Rescue, Port Tender, Anti-pollution, Law Enforcement, Industrial Vessel, MODU (Offshore Drilling), Research Vessel).

3 Preprocessing and Imbalance Mitigation

3.1 Splits and leakage control

We perform stratified splits into train/validation/test with a two-stage 80/20 then 80/20 procedure. Although the dataset is already ship-level, MMSI is retained as a grouping key for future extensions where multiple records per ship may exist.

3.2 Imputation and scaling

Missing values are imputed by the **median** (robust to outliers). Features are standardized with **z-score scaling** fit on training data only and applied to validation/test to avoid leakage.

3.3 Hybrid imbalance strategy

The class distribution is highly skewed (few dominant classes, many rare ones) [?, ?]. We combine:

- **Capped undersampling:** for any class with count $> CAP =$

3000, randomly downsample to CAP.

- **Threshold oversampling:** if count $< TH_VERY_RARE = 120$, oversample to $TGT_VERY_RARE = 350$; if count $< TH_RARE = 220$, oversample to $TGT_RARE = 280$.

Finally, we compute **class weights** from the post-undersampling / pre-oversampling distribution using a median-count ratio, then clip to stabilize optimization.

4 Model

We use a residual multi-layer perceptron (MLP) for tabular data:

$$\mathbf{x} \in \mathbb{R}^{11} \rightarrow [256, 256, 128, 128] \rightarrow \text{softmax}(17).$$

Each block follows $Dense(ReLU + L2) \rightarrow BatchNorm \rightarrow Dropout$, with dropout rates (0.35, 0.35, 0.30, 0.30) and L2 regularization 10^{-5} . When dimensions match, we add a **residual skip connection** to stabilize gradients and allow refinement of representations. The output layer uses **softmax** with categorical cross-entropy (optionally with label smoothing).

5 Training and Evaluation

We train with Adam (initial learning rate 10^{-3}), batch size 1024, up to 300 epochs. Two callbacks control convergence: EarlyStopping on validation loss (patience 25, restore best), and ReduceLROnPlateau (factor 0.5, patience 8, min LR 10^{-5}). We report test accuracy and class-wise precision/recall/F1 and inspect a row-normalized confusion matrix to localize confusions among similar classes.

6 Results

6.1 Overall performance

On the held-out test set, the proposed baseline achieves a test loss of 0.5051 and an accuracy of 0.8149. Macro-averaged performance is strong (macro F1 = 0.8028), indicating that the classifier remains reasonably balanced across classes, while the weighted F1 = 0.8139 confirms that performance is not driven solely by majority classes.

6.2 Per-class behavior and confusion patterns

Most classes exhibit high recall (row-wise correctness in the confusion matrix), notably *Search & Rescue* (recall 0.9940, F1 0.9763) and *Sailing & Pleasure* (recall 0.9300, F1 0.9502). However, several systematic confusions emerge.

Cargo vs Tanker is the dominant failure mode. Cargo is misclassified as Tanker in 85/400 cases (21.25%), and Tanker is misclassified as Cargo in 86/400 cases (21.5%). This near-symmetric swap suggests that the current feature set (mainly geometry and mean speed) is insufficient to separate two large classes with overlapping size and operating-speed profiles.

Fishing is the weakest medium-support class. Fishing attains precision 0.5472 and recall 0.5088 (F1 0.5273). The

confusion matrix shows that Fishing vessels are often predicted as *Tug* ($30/114 = 26.3\%$) and, to a lesser extent, as *Cargo* ($10/114 = 8.8\%$). Conversely, many non-Fishing vessels are predicted as Fishing (low precision), primarily *Passenger* (18 cases) and *Tug* (18 cases), consistent with overlapping geometric footprints and speed statistics, and possibly amplified by targeted rebalancing for Fishing.

Industrial Vessel is the hardest minority class. It has the lowest recall (0.4000) and the lowest F1 (0.4615), with only 15 test samples. Errors are dispersed across semantically related categories (e.g., *Dredging/Underwater ops* and *Sailing & Pleasure*), indicating both data scarcity and within-class heterogeneity (“Industrial Vessel” often acts as a broad catch-all label).

Diving ops shows a precision–recall trade-off. Despite a decent recall (0.7727), its precision is lower (0.6071), meaning the model predicts Diving ops too often. False positives mainly come from *Passenger* (6 cases) and *Sailing & Pleasure* (4 cases), which is plausible when only coarse speed and shape features are available.

Law Enforcement has very high recall but reduced precision. Law Enforcement reaches recall 0.9821 but precision 0.7857 because *Military ops* is frequently predicted as Law Enforcement

(10 cases), showing that these two operational classes partially overlap under the chosen features.

7 Conclusion

We presented a compact, reproducible vessel-type classification pipeline from NOAA AIS data using only 11 ship-level features and a residual feed-forward neural network. A hybrid imbalance strategy (sampling + weighted loss) enables competitive performance in a 17-class setting, producing a strong, lightweight baseline suitable for extension to richer AIS behavior modeling and anomaly detection.

References (placeholders)

References

- [1] R. Garg, A. P. Kapoor, V. H. Hariya, J. Acharya, and A. Mundra, “Systematic Review of AIS-Based Trajectory Prediction Models,” *[Journal/Conference]*, 2024.
- [2] H. Kim *et al.*, “Vessel Trajectory Classification via Transfer Learning with Deep Convolutional Neural Networks,” *[Journal/Conference]*, 2024.
- [3] P. Last *et al.*, “Comprehensive Analysis of AIS Data in Regard to Vessel Movement Prediction,” *[Journal/Conference]*, 2014.